Systems Analysis and Design

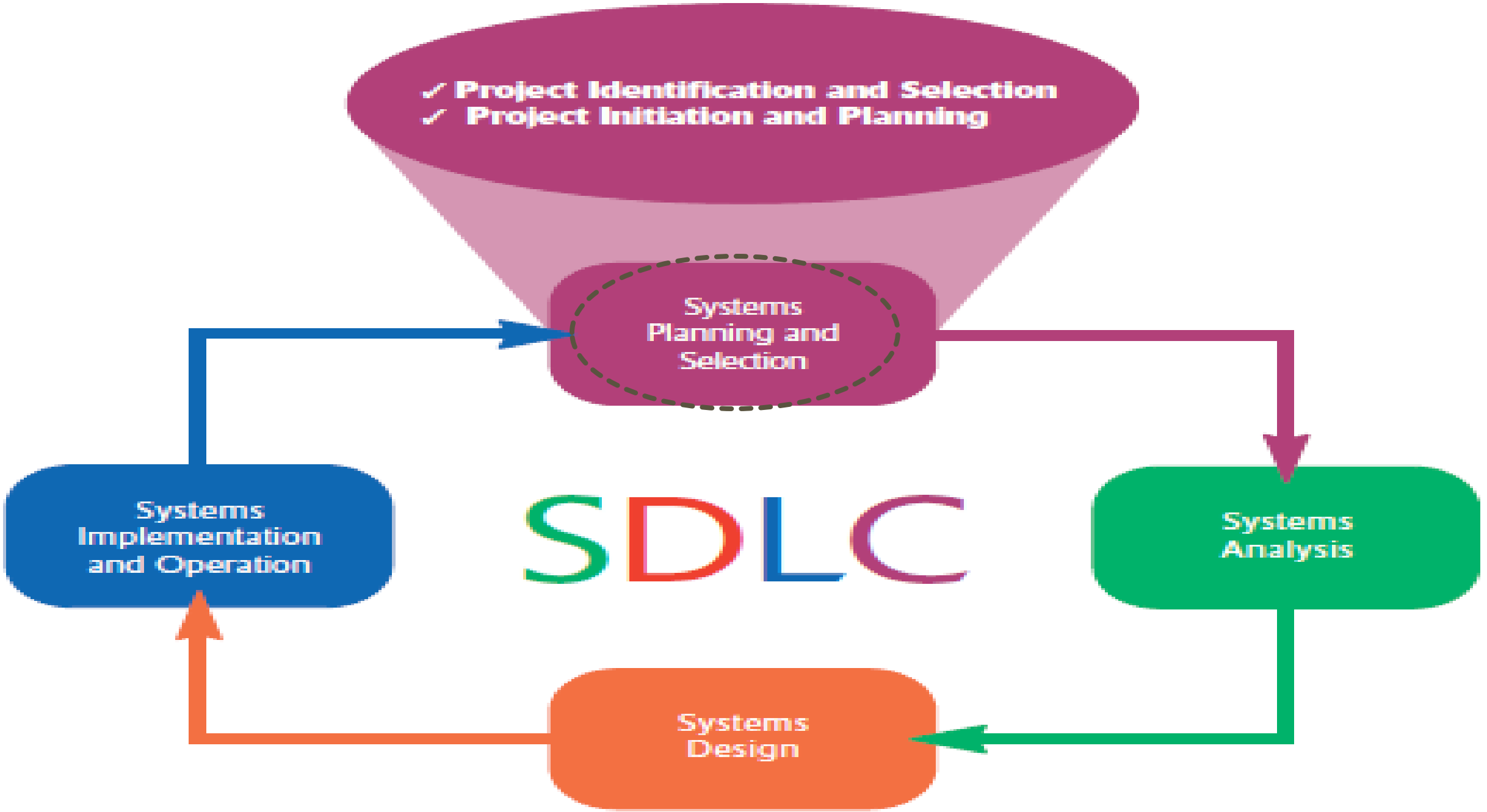


Dr. Alaa Eldin Abdallah Yassin

Information system department

Lecturer 3 :
Chapter 3&4- part1
Systems Planning and Selection

Dr. Alaa Eldin Abdallah Yassin


✓ Project Identification and Selection
✓ Project Initiation and Planning
Systems Planning and Selection
Systems Analysis
Systems Design
Systems Implementation and Operation
SDLC

Learning Objectives

- Describe the steps involved in the planning process.
- Explain the need for and the contents of a Project Scope Statement and Baseline Project Plan.
- List and describe various methods for assessing project feasibility.
- Describe the **differences** between tangible and intangible benefits and costs and between one-time vs recurring benefits and costs*.
- Perform cost-benefit analysis and describe what is meant by the time value of money, present value, discount rate, net present value.*

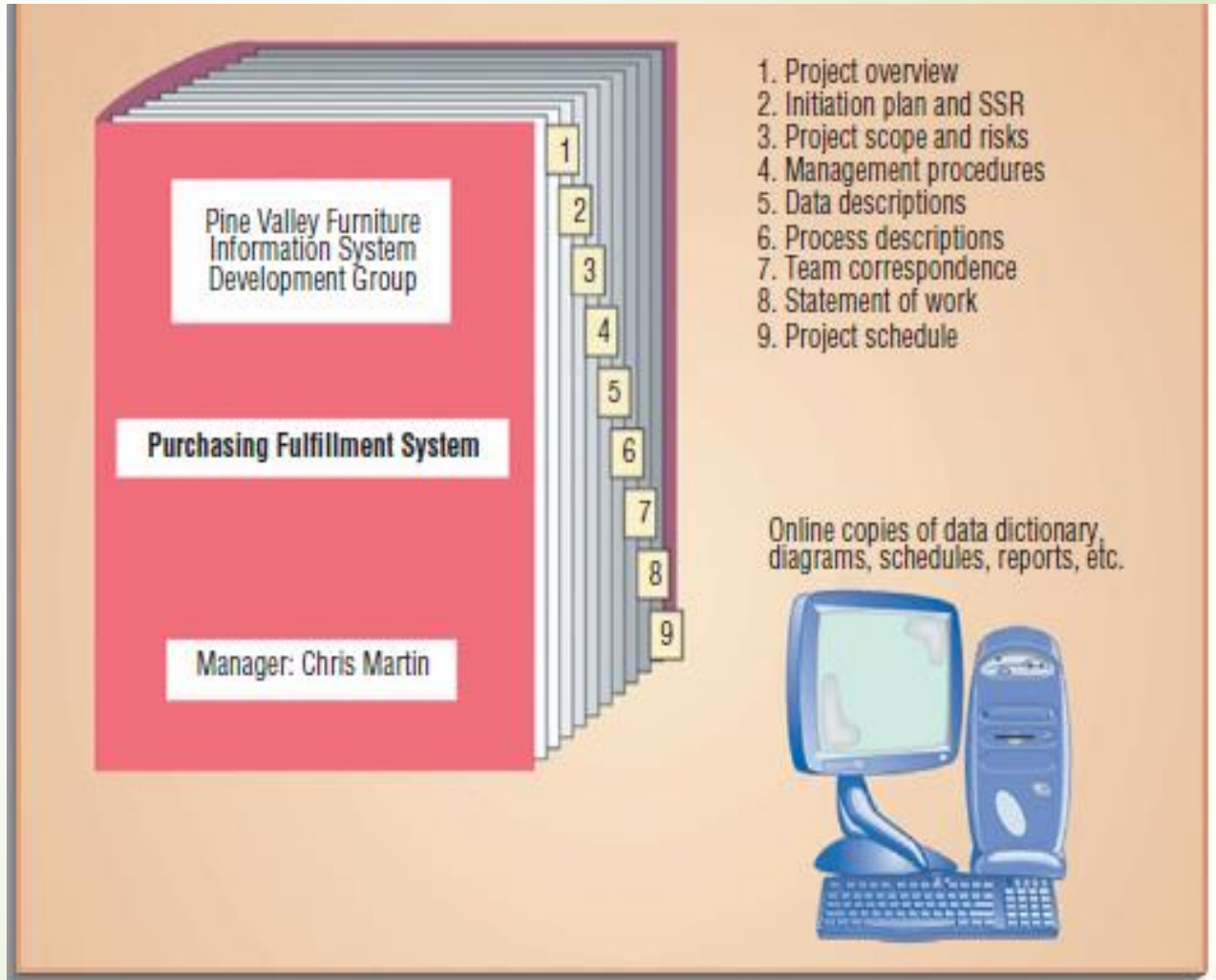
Initiating and Planning projects

The Process of Initiating IS Development Projects

- Project initiation focuses on activities designed to assist in organizing a team to conduct project planning.
- The main steps of Project initiation :
  - Establishing the Project Initiation Team.
  - Establishing a Relationship with the Customer.
  - Establishing the Project Initiation Plan.
  - Establishing the Project Management Environment and Project Workbook.
  - Developing the Project Charter.

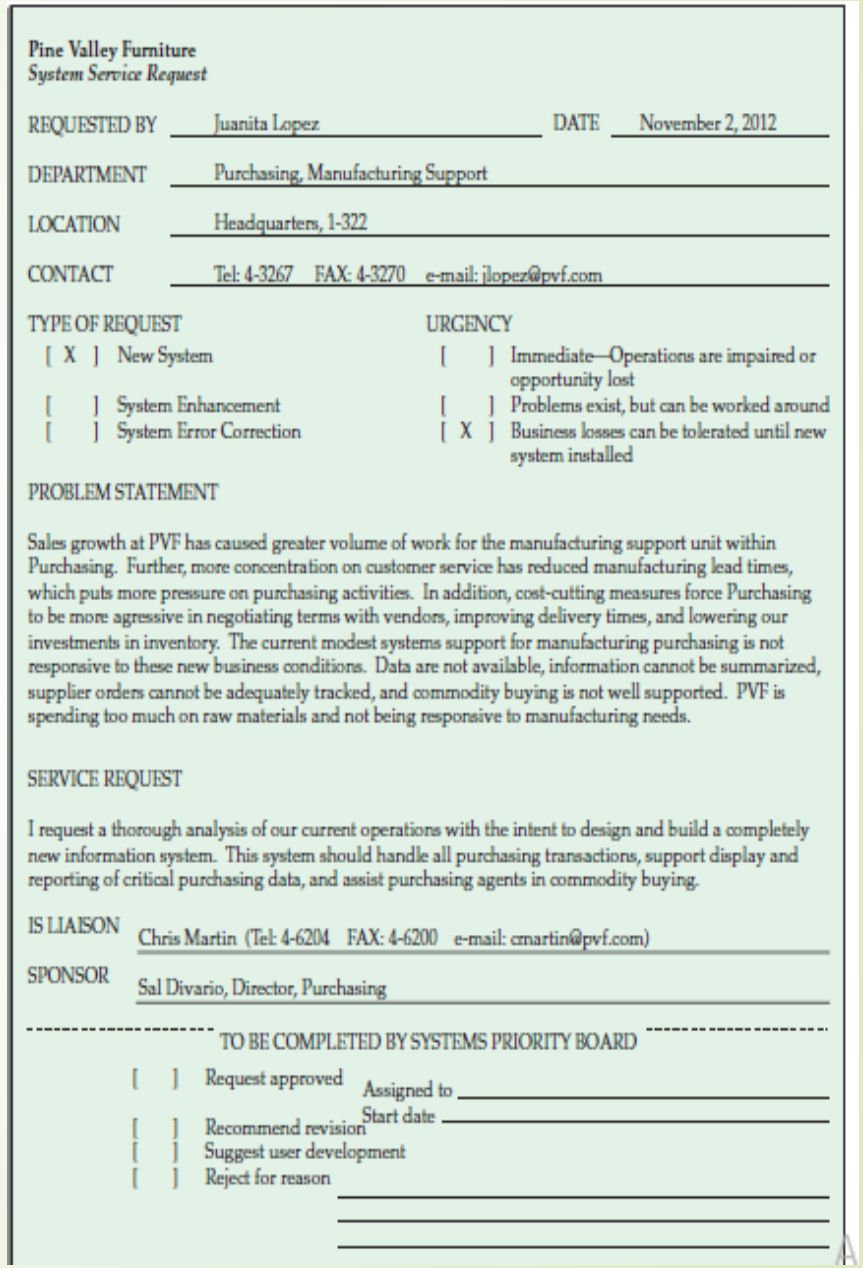
The Process of Initiating IS Development Projects… Project Workbook.

- The contents of workbook
  1. Project overview
  2. Initiation plan and SSR*
  3. Project scope and risks
  4. Management procedures
  5. Data descriptions
  6. Process descriptions
  7. Team correspondence
  8. Statement of work.
  9. Project schedule



SSR: System Service Request example

- SSR for purchasing fulfillment :
- with name and contact information of the person requesting the system,
- a statement of the problem,
- and the name and contact information of the liaison and sponsor.
- Type of request (new system or enhanced one).
- …



Pine Valley Furniture
System Service Request

REQUESTED BY Juanita Lopez DATE November 2, 2012

DEPARTMENT Purchasing, Manufacturing Support

LOCATION Headquarters, 1-322

CONTACT Tel: 4-3267 FAX: 4-3270 e-mail: jlopez@pvf.com

TYPE OF REQUEST
[X] New System
[ ] System Enhancement
[ ] System Error Correction

URGENCY
[ ] Immediate—Operations are impaired or opportunity lost
[ ] Problems exist, but can be worked around
[X] Business losses can be tolerated until new system installed

PROBLEM STATEMENT

Sales growth at PVF has caused greater volume of work for the manufacturing support unit within Purchasing. Further, more concentration on customer service has reduced manufacturing lead times, which puts more pressure on purchasing activities. In addition, cost-cutting measures force Purchasing to be more agressive in negotiating terms with vendors, improving delivery times, and lowering our investments in inventory. The current modest systems support for manufacturing purchasing is not responsive to these new business conditions. Data are not available, information cannot be summarized, supplier orders cannot be adequately tracked, and commodity buying is not well supported. PVF is spending too much on raw materials and not being responsive to manufacturing needs.

SERVICE REQUEST

I request a thorough analysis of our current operations with the intent to design and build a completely new information system. This system should handle all purchasing transactions, support display and reporting of critical purchasing data, and assist purchasing agents in commodity buying.

IS LIAISON Chris Martin (Tel: 4-6204 FAX: 4-6200 e-mail: cmartin@pvf.com)

SPONSOR Sal Divario, Director, Purchasing

TO BE COMPLETED BY SYSTEMS PRIORITY BOARD

[ ] Request approved Assigned to
Start date
[ ] Recommend revision
[ ] Suggest user development
[ ] Reject for reason

The Process of Initiating IS Development Projects…

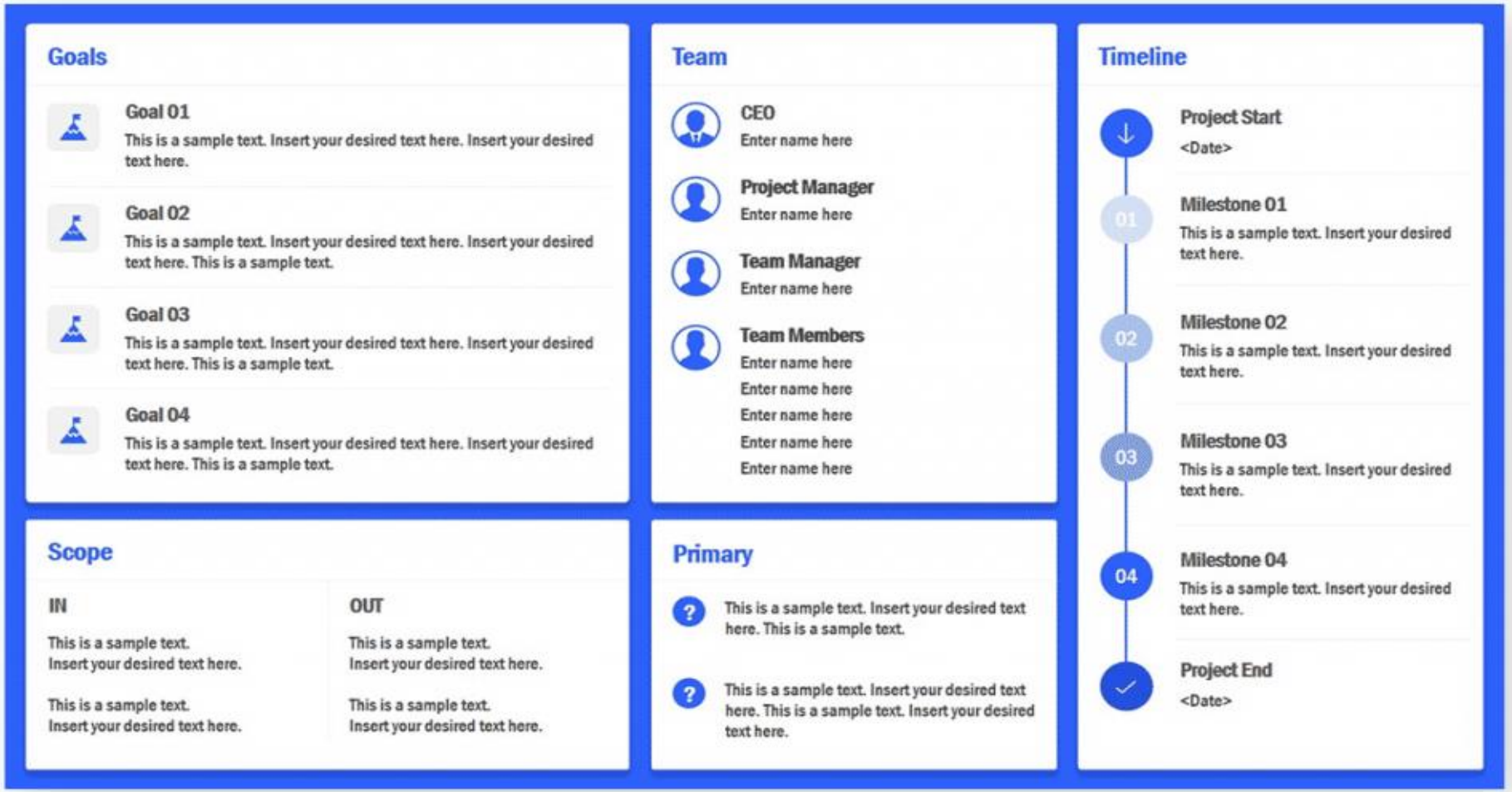
Project Charter

- The key activity of project initiation is the development of the project charter.
- Q: What is a Project Charter?
  - A short document that is prepared for both **internal** and **external** stakeholders.
  - Provides a high-level overview of the project (provides a bird-eye overview)
  - Useful communication tool that helps to assure that the organizations and other stakeholders understand the initiation of a project
- Q: Who is responsible for performing this function?
  - that's the job of the project initiator — a project leader, manager, or another stakeholder who's going to be in charge of the initiative

The Process of Initiating and Planning IS Development Projects…

- Q: Mention the contents of project charter.
- A project charter typically contains:
  - Project title and date of authorization.
  - Project manager name and contact information
  - Customer name and contact information
  - Project start and completion dates
  - Key stakeholders, project role, and responsibilities
  - Project objectives and description
  - Key assumptions or approach
  - Signature section for key stakeholders

Template of project charter



Template of project charter

Pine Valley Furniture

Project Charter **Prepared:** November 2, 2012

Project Name:	Customer Tracking System
Project Manager:	Jim Woo (jwoo@pvf.com)
Customer:	**Marketing**
Project Sponsor:	Jackie Judson (jjudson@pvf.com)
Project Start/End (projected):	10/2/12–2/1/13

Project Overview:

This project will implement a customer tracking system for the marketing department. The purpose of this system is to automate the . . . to save employee time, reduce errors, have more timely information . . .

Objectives:

- Minimize data entry errors
- Provide more timely information
- . . .

Key Assumptions:

- System will be built in-house
- Interface will be a Web-browser
- System will access customer database
- . . .

Stakeholders and Responsibilities

Stakeholder	Role	Responsibility	Signatures
Jackie Judson	VP Marketing	Project Vision, Resources	Jackie Judson
Alex Datta	CIO	Monitoring, Resources	Alex Datta
Jim Woo	Project Manager	Plan, Monitor, Execute Project	Jim Woo
James Jordan	Director of Sales	System Functionality	James Jordan
Mary Shide	VP Human Resources	Staff Assignments	Mary Shide

مثال عملي: نظام حجز تذاكر القطارات

Project Title: Railway Ticket Reservation System

العنصر	المثال العملي
Project Name	Railway Ticket Reservation System
Project Sponsor	Ministry of Transport
Project Manager	Ahmed Ali
Project Start Date	October 2026 1
Expected End Date	March 2027 31
Project Purpose	تطوير نظام إلكتروني يسمح للمستخدمين بالبحث عن الرحلات وحجز تذاكر القطارات ودفع قيمتها إلكترونيًا.
Business Need	تقليل الازدحام في مكاتب الحجز، وتقليل الأخطاء اليدوية، وتحسين خدمة العملاء.
Project Objectives	1. إنشاء نظام للحجز الإلكتروني. 2. إتاحة الدفع الإلكتروني. 3. عرض الرحلات والمقاعد المتاحة. 4. إصدار التذكرة إلكترونيًا. 5. توفير تقارير للإدارة.
Major Deliverables	Website. Database. Admin Dashboard. Mobile-friendly interface. Payment module. Electronic ticket.

Main Stakeholders	Passengers، Railway Management، Ticket Officers، IT Department، Payment Service Provider.
High-Level Requirements	تسجيل المستخدم، البحث عن الرحلات، اختيار المقعد، الحجز، الدفع، إلغاء الحجز، إصدار التذكرة.
Estimated Budget	EGP 500,000
Major Risks	فشل النظام، مشاكل أمن البيانات، تأخر التكامل مع بوابة الدفع، زيادة عدد المستخدمين.
Constraints	الميزانية محدودة، مدة المشروع 6 أشهر، ضرورة الالتزام بسياسات أمن المعلومات.
Assumptions	توفر بيانات الرحلات، توفر خوادم مناسبة، تعاون إدارة السكك الحديدية مع فريق التطوير.
Success Criteria	تشغيل النظام بنجاح، قدرة المستخدم على إتمام الحجز خلال دقائق، انخفاض الاعتماد على الحجز اليدوي، رضا المستخدمين.
Project Approval	Project Sponsor approval is required before project execution.

Project charter

PROJECT CHARTER

Project: Railway Ticket Reservation System

1. Project Purpose:

To develop an online railway ticket reservation system that allows passengers to search for trains, select seats, make payments, and receive electronic tickets.

2. Project Objectives:

- Reduce waiting time at railway stations.
- Reduce manual booking errors.
- Provide online payment.
- Improve customer satisfaction.

3. Major Deliverables:

- Web-based reservation system.
- Database.
- Payment module.
- Admin dashboard.
- Electronic ticket system.

4. Stakeholders:

Passengers, Railway Management, IT Department, Ticket Officers.

5. Major Risks:

Security attacks, system failure, payment gateway problems, and schedule delays.

6. Budget:

500,000 EGP.

Project charter…

4. Stakeholders:

Passengers, Railway Management, IT Department, Ticket Officers.

5. Major Risks:

Security attacks, system failure, payment gateway problems, and schedule delays.

6. Budget:

500,000 EGP.

7. Schedule:

October 2026 – March 2027.

8. Project Manager:

Ahmed Ali.

9. Approval:

The project sponsor authorizes the project manager to proceed with project planning and execution.

The Process of Planning IS Development Projects…

- Q: What is the key activity of project planning?
  - The key activity of project planning is the process of defining clear, discrete activities and the work needed to complete each activity within a single project.

- Q: What is the objective of the project planning process?
  - The objective of the project planning process is the development of a Baseline Project Plan (**BPP**)* and the Project Scope Statement (**PSS**).

Elements of Project Planning

Q: Mention the different elements of project planning ?

1. Describe project scope, alternatives, feasibility.
2. Divide project into tasks.
3. Estimate resource requirements and create resource plan.
4. Develop preliminary schedule.
5. Develop communication plan.
6. Determine standards and procedures.
7. Identify and assess risk.
8. Create preliminary budget.
9. Develop a statement of work.
10. Set baseline project plan.

The Deliverables and Outcomes…

- **Project Scope Statement (PSS)**
  - A document prepared for the customer.
  - Describes what the project will deliver.
  - Outlines at a high level all work required to complete the project.

The Deliverables and Outcomes

- **Business Case** : A written report that outlines :
  - The justification for an information system.
  - Economic benefits and costs
  - and the technical and organizational feasibility of the proposed system.

The Deliverables and Outcomes…

- **Baseline Project Plan (BPP)**
  - A major outcome and deliverable from the **PIP** (project initiation and planning) phase.
  - Contains the best estimate of a project's scope, benefits, costs, risks, and resource requirements

Elements of Project Planning: project scope



- What is project scope?
  - Project Scope is the description of all the work that needs to be done to create deliverables and achieve the project objective.
  - Project scope is a detailed outline of all aspects of a project, including all related activities, resources, timelines, and deliverables, as well as the project's boundaries.
  - A project scope also outlines key stakeholders, processes, assumptions, and constraints, as well as what the project is about, what is included, and what isn't.
  - All of this essential information is documented in a *scope statement*.

Elements of Project Planning: project scope…

- Important of project scope
  - To create an accurate project budget or schedule you need to identify 100% of the project scope.
  - Risk Management will depend on the clarity of the scope of work.
  - Quality Management is simply a part of the scope.

Elements of Project Planning: project scope…

- The best tools to describe project scope are Project Scope Statement,
- **Project Scope Statement**
  - The project scope statement is a key document that provides all stakeholders with a clear understanding of why the project was initiated and defines its key goals.
  - Describes what the project will deliver.
  - Outlines at a high level all work required to complete the project.

Project Scope example
Customer Tracking System

- **Introduction**
  - Sales growth has outpaced the marketing department's ability to track and forecast customer buying trends accurately. An improved method for performing this process must be found in order to reach company objectives.
- **Project Scope**
  - A new information system will be constructed that will **collect** all customer purchasing activity, support display and reporting of sales information, aggregate data, and show trends in order to assist marketing personnel in understanding dynamic market conditions.
  - The project will follow PVF's systems development life cycle.

Project scope example
Customer Tracking System…..

- **Project Deliverables**
  - Customer tracking system analysis and design
  - Customer tracking system programs
  - Customer tracking documentation
  - Training procedures

Project scope example:
content marketing project

Introduction

This content marketing project is being undertaken for X company for the purpose of creating an article to be posted on their site to create brand awareness.

Project Scope

This project will include research, content strategy, writing the article, and publishing it on X's website under the X blog. It will also include sharing the article on social media for the month of May 2022. All activities will be conducted by Mohamed Hussein of A company.

Project scope example:
content marketing project…..

Project Deliverables

Project deliverables will include one well-researched written article of up to 1,000 words to be delivered by email to Mohamed@X.com.

Project Acceptance Criteria

Jane at X company will review and approve the final article version before publishing

Project Exclusions

This project will not include payment to external vendors for research or outsourced services

Project Constraints

Constraints may include communication delays, changes in scope, or technical difficulties.

Elements of Project Planning: Assessing Project Feasibility



- Most information systems projects have budgets and deadlines.
- Assessing project feasibility is a required task that can be a large undertaking because it requires the systems analyst, to evaluate a wide range of factors.
- Although the specifics of a given project will dictate which factors are most important
- Most feasibility factors fall into the following categories:
  1. Economic
  2. Technical
  3. Operational
  4. Scheduling
  5. Legal and contractual
  6. Political

1. Assessing economic feasibility

- What is the economic feasibility ?
  - Economic feasibility is a process of identifying the financial benefits and costs associated with a development project
  - Often referred to as a *cost-benefit analysis*
  - Project is reviewed after each SDLC phase in order to decide whether to continue, redirect, or kill a project.

1. Assessing economic feasibility….

One time cost

- cost associated with project start up and development or system start up
- These costs encompass activities such as:
  - Systems development,
  - New hardware and software purchases,
  - User training,
  - Site preparation.

Recurring cost:

- a cost resulting from the ongoing evolution and use of a system
- Examples of these costs include
  - Application software maintenance,
  - Incremental data storage expenses,
  - Incremental communications,
  - New software and hardware leases, a
  - Supplies and other expenses (i.e. paper, forms, data center personnel).

1. Assessing economic feasibility….

One –time cost(example)

ONE-TIME COSTS WORKSHEET
Customer Tracking System Project

	Year 0
A. Development costs	$20,000
B. New hardware	15,000
C. New (purchased) software, if any	
1. Packaged applications software	5,000
2. Other _____	0
D. User training	2,500
E. Site preparation	0
F. Other _____	0
TOTAL One-Time Costs	**$42,500**

Recurring cost(example)

RECURRING COSTS WORKSHEET
Customer Tracking System Project

	Year 1 through 5
A. Application software maintenance	$25,000
B. Incremental data storage required: 20 GB × $50 (estimated cost/GB = $50)	1,000
C. Incremental communications (lines, messages, …)	2,000
D. New software or hardware leases	0
E. Supplies	500
F. Other _____	0
TOTAL Recurring Costs	**$28,500**

1. Assessing economic feasibility….

Project Benefits

benefits are both tangible and intangible.

- Tangible benefits refer to items that can be measured in dollars and with certainty.

Examples include:

- Increased Revenue
- Increased sales
- Resource Cost Savings
- Higher profit margins.

1. Assessing economic feasibility…

- Intangible benefits are benefits derived from the creation of an information system that cannot be easily measured in dollars or with certainty
- May have direct organizational benefits, such as the
  - improvement of employee moral: تحسين فى أخلاقيات و حالة الموظفين
  - Brand Equity : حقوق الملكية
- May have broader societal implications, such as the reduction of waste creation or resource consumption

1. Assessing economic feasibility….

Tangible cost

- a cost associated with an information system that can be measured in dollars and with certainty
- IS development tangible costs include:
  - Hardware costs,
  - Labor costs, or
  - Operational costs including employee training and building renovations.

Intangible cost

- a cost associated with an information system that cannot be easily measured in terms of dollars or with certainty
- Intangible costs can include:
  - Loss of customer goodwill,
  - Employee morale, or
  - Operational inefficiency.

1. Assessing economic feasibility….

TANGIBLE BENEFITS WORKSHEET
Customer Tracking System Project

	Year 1 through 5
A. Cost reduction or avoidance	$ 4,500
B. Error reduction	2,500
C. Increased flexibility	7,500
D. Increased speed of activity	10,500
E. Improvement in management planning or control	25,000
F. Other _____	0
TOTAL Tangible Benefits	**$50,000**

1. Assessing economic feasibility….

The Time Value of Money (TVM)

- The time value of money (TVM) is the concept that money you have now is worth more than the identical sum in the future due to its potential earning capacity. *
- The process of comparing present cash outlays to future expected returns
- Use discount rate to determine present value of cash outlays and receipts

$$PV = FV \times \frac{1}{(1+i)^n}$$

Where, where PV is the present value and FV future value when i is the discount rate.

discount rate i: The interest rate used to convert future benefits and costs to their present value in year 1

1. Assessing economic feasibility….

- Net Present Value (NPV)
  - The conversion of a stream of future benefits less future costs to their equivalent benefits and costs in year 1, at the beginning of the project or program.
  - NPV =sum of all PVs across years

1. Assessing economic feasibility….

- Why is the Time Value of Money important?
  - Time value of money is very important because it can help guide investment decisions.
- For instance, suppose an investor can choose between two projects: Project A and Project B. Both projects have identical descriptions except that Project A promises a $1 million cash payout in year 1, whereas Project B offers a $1 million cash payout in year 5. If the investor did not understand the time value of money, they might believe that these two projects are equally attractive. In fact, however, time of money dictates that Project A is more attractive than Project B because its $1 million payout has a higher present value.

2. Assessing Technical Feasibility

- What is Technical feasibility?
  - Technical feasibility: is a process of assessing the development organization's ability to construct a proposed system.
- The potential consequences of **not assessing** and managing risks can include the following:
  - Failure to attain expected benefits from the project,
  - Inaccurate project cost estimates,
  - Inaccurate project duration estimates,
  - Failure to achieve adequate system performance levels, and
  - Failure to adequately integrate the new system with existing hardware, software, or organizational procedures

3. Assessing Operational feasibility

- Operational feasibility
  - Does the proposed system solve problems or take advantage of opportunities?

4. Assessing Scheduling feasibility

- Scheduling feasibility
  - Can the project time frame and completion dates meet organizational deadlines?

5. Assessing Legal feasibility

- Legal and Contractual feasibility
  - What are legal and contractual ramifications of the proposed system development project?

6. Assessing Political feasibility

- Political feasibility
  - How do key stakeholders view the proposed system?

Review Questions

1. What is the Project Charter? and Who is responsible for performing this function?
2. What mean by SSR in project initiation process?
3. Mention the contents of project charter.
4. What is the key activity of project planning?
5. What is the objective of the project planning process?
6. Mention the different elements of project planning ?
7. Talk in short notes about : BBP, PPS and business case.
8. What is project scope? and its importance ?
9. Project Scope Statement is the best tool to describe project scope, how?
10. What is the economic feasibility ?
11. Why is the Time Value of Money important?
12. What is Technical feasibility?

internal and external stakeholders

- Internal : داخلى
  - أشخاص موجودون داخل المنظمة و يتفاعلون مباشرة مع المشروع و يتأثرون به
  - أمثلة :
    - مدير المشروع
    - مدير التطوير
    - الدعم الفنى
    - إدارة الشركة
- External : خارجى
  - أشخاص خارص المنظمة و لكن لهم تأثير على المشورع او يتأثرون بنتائج المشروع
  - أمثلة :
    - العملاء
    - المستخدم النهائى
    - الجهات التنظيمية


Thank You!